



Efficient Object Detection Framework for Indian Road Scenarios Under Foggy Conditions Using YOLOv8

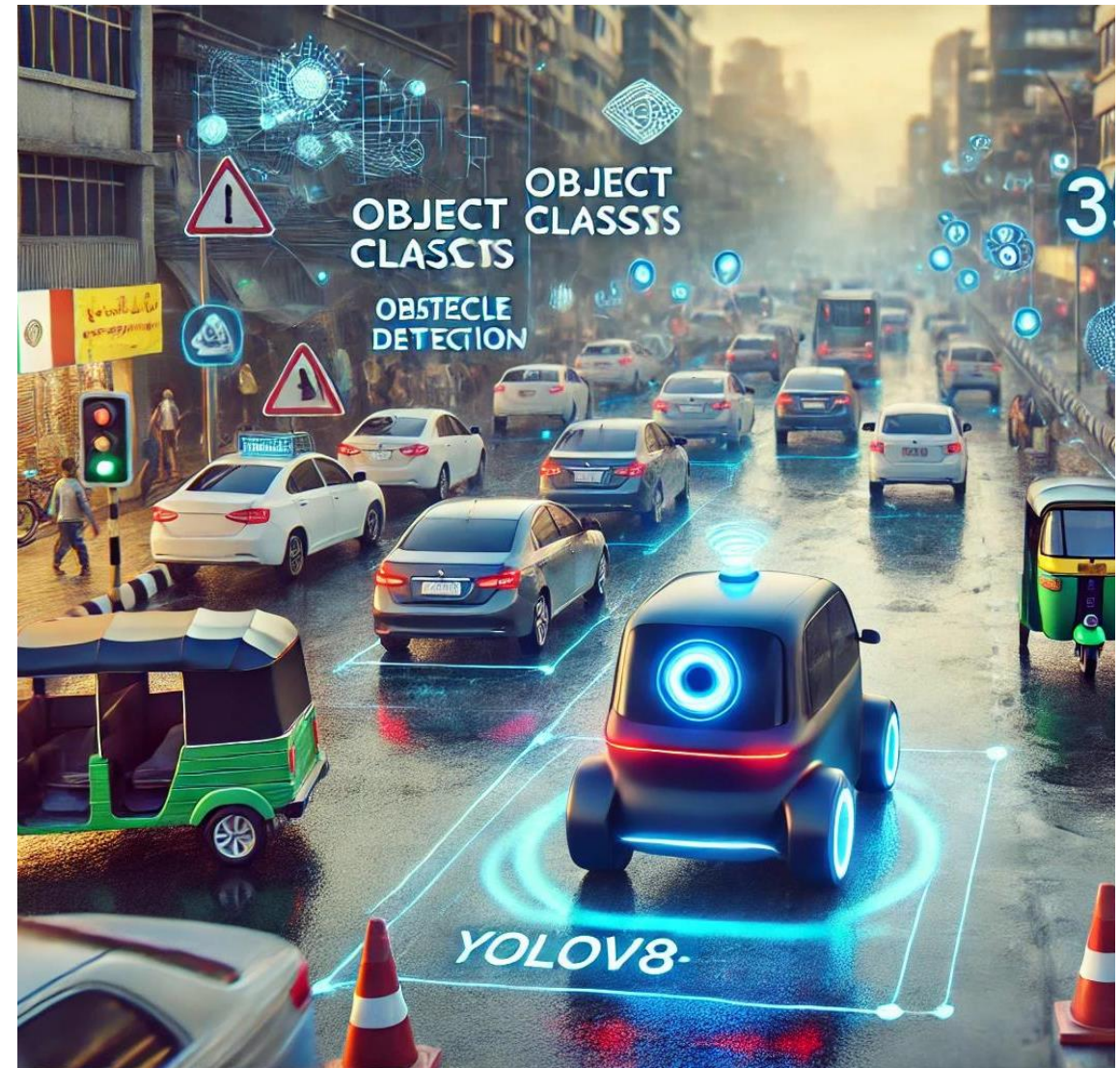
	NAME	ROLL NO	SRN
1.	ROHAN KULKARNI	27	02FE22BCS083
2.	ANURAG CHOUGULE	09	02FE22BCS021
3.	MRUNALI BENNALKAR	16	02FE22BCS055
4.	REVATI BELAGALI	26	O2FE22BCS082

TEAM NO-1

Under the Guidance of : Dr. Prema T. Akkasaligar

Introduction

This project focuses on real-time obstacle detection and classification using the YOLOv8 architecture, optimized for Indian road conditions. It identifies 35 distinct object classes, including vehicles, pedestrians, and traffic signs, and is trained on the Indian Driving Dataset (IDD) to handle challenges like dense traffic and erratic driver behaviour. The system ensures high accuracy and precision under varying weather and lighting conditions. Designed to be scalable and memory-efficient, it is well-suited for deployment in resource-constrained environments like autonomous vehicles, ensuring reliable and efficient operation.



Empathy

It captures the vulnerability of pedestrians amidst the chaos of bustling roads, emphasizing the critical need for safety. It reflects the emotional and physical toll accidents impose on individuals and families, urging us to prioritize preventive measures. Every life on the road deserves care and protection, and such moments remind us of our shared responsibility to create safer environments. By addressing these challenges with compassion, we can ensure a future where everyone feels secure and valued on their journey.



Background & Motivation

Indian roads are known for their complexity, with dense traffic, erratic driving behaviour, and diverse weather conditions. Traditional obstacle detection systems often struggle to handle these challenges, making it difficult for autonomous vehicles to navigate safely. These conditions require a robust solution capable of real-time object detection and classification, especially for varied road elements such as vehicles, pedestrians, and infrastructure.

The motivation behind this project is to enhance road safety and navigation efficiency in challenging environments like Indian roads. By leveraging YOLOv8 for real-time, high-accuracy object detection, the project aims to address the unique challenges of Indian traffic. The goal is to create a scalable, memory-efficient system suitable for deployment in resource-constrained autonomous vehicles, ultimately contributing to safer, smarter transportation.

Literature Review

Paper Title	Methodology/Implementation	Data set Description	Performance Parameter	Limitations/Disadvantages	Gaps Identified
1.Lane and Object Detection using YOLO for Indian Roads Scenario	Implemented YOLO for lane/object detection on Indian roads; real-time, weather-sensitive.	The dataset has Indian road images for training YOLO in ADAS.	"High detection accuracy, real-time speed, robust, but reduced in severe weather.	Reduced accuracy in heavy rain, dust storms, poor lane markings, complex traffic	The system struggles in extreme weather and inconsistent or missing lanes.
2."Object-based image retrieval and detection for surveillance video using YOLOv8."	YOLOv5 outperformed YOLOv8 in detecting potholes and traffic signs	The dataset features Indian road images with potholes and traffic signs.	Used precision, recall, and mAP to compare YOLOv5 and YOLOv8	Faster performance, but lower precision and recall than YOLOv5.	YOLOv8 underperformed in complex Indian road detection accuracy

Literature Review

3.Comparative analysis of YOLOv5 and YOLOv8 for detecting potholes and traffic signs on Indian roads	Proposed framework enhances object detection, balancing accuracy and overhead.	ImageNet VID and VisDrone-VID2019 are annotated video datasets for detection	Evaluate model using mAP for accuracy and fps for speed.	Pre-training for video prediction adds computational overhead, affecting scalability.	Need better scalability, robustness, and optimized pre-training for efficiency.
4.Video Representation Learning through Stochastic Video Prediction for Online Object Detection	Faster R-CNN excels in accuracy; SSD better in localization.	Open Image Dataset V7 aids object detection with diverse annotations.	Evaluate object detection with mAP and recall metrics.	Faster R-CNN has lower recall and higher computational complexity.	Improve object localization, optimize model efficiency, and explore real-time applications

Literature Review

5.Object Detection in Autonomous Vehicles: A Performance Analysis using SSD and Faster R-CNN	System detects moving objects accurately but struggles with complex backgrounds.	Dataset includes annotated surveillance videos of humans and cars.	K-NN classifier achieved high accuracy in surveillance object classification.	Limited effectiveness in dynamic multimodal backgrounds affects detection accuracy.	Model lacks adaptability to dynamic environments; needs further research.
6.Detection and classification of moving objects Using background subtraction and KNN	The system uses background subtraction and KNN for object detection.	Dataset includes video frames for detecting humans and vehicles.	Model performance evaluated by accuracy, precision, recall, and F1-score.	Effectiveness decreases in complex, varied environments and interactions.	System struggles with diverse conditions and needs better algorithms.

Literature Review

7.Benchmark for Road Abandoned Object Detection (RAOD) using video surveillance	Yajun Xu's method improves segmentation but complicates real-time implementation .	Annotated highway surveillance footage for training segmentation models.	The Dice score evaluates segmentation accuracy alongside precision and recall.	Increased complexity may hinder real-time deployment and segmentation accuracy.	Study lacks evaluation in extreme weather and traffic scenarios.
8.Efficient Object Detection and Classification using Enhanced Moving Object Detection(EMOD) and Manta Ray CNN	EMOD uses optimized CNN for accurate, real-time moving object detection.	EMOD uses annotated motion videos for robust object detection training.	EMOD performance is evaluated using sensitivity, specificity, and accuracy.	EMOD's complexity limits real-time processing in challenging environments.	EMOD requires broader evaluation and lower resource dependency for deployment.

Literature Review

9.YOLOv8 model for object and pedestrian detection in foggy weather conditions	YOLOv8 optimizes object detection but faces tuning complexity challenges.	Dataset includes foggy images with annotated objects for YOLOv8.	Model performance is measured using precision, recall, and mAP50	Hyperparameter tuning is complex, limiting deployment on resource constrained devices.	Study gaps include weather evaluation and real-time processing limitations.
10.Video classification using CNN and LSTM for object detection	Srinath and Baydeti achieved 92.86% accuracy using CNNs and LSTMs.	Dataset includes UCF101 videos for evaluating action recognition algorithms.	Model achieved 92.86% accuracy on UCF101, outperforming traditional methods.	Increased complexity risks overfitting, complicating training and deployment.	Efficient architectures needed to balance accuracy and computational efficiency.

Literature Review

11.Multi object road waste detection and classification using binocular camera and CNN	Guo and Chen's system uses binocular vision for waste detection.	The system evaluated waste objects under diverse road conditions.	Improved detection accuracy and speed, but affected by environment.	Environmental factors impact stereo matching accuracy and system performance.	Study overlooks stereo matching improvements and additional sensor integration
12.Ontology based deep learning model for object detection using YOLOv8	Adegun et al. combined YOLOv8 and ontology for satellite analysis.	Dataset features urban scenes for testing model effectiveness in environments	Model achieved 68% precision and 60% recall in object detection	Ontology model complexity limits scalability and adaptability in urban settings.	Further research is needed to simplify ontology construction and scalability.

Course project objectives

1. Develop a deep learning-based system for real-time object detection and classification on Indian roads, addressing traffic and weather challenges.
2. Implement YOLOv8 Nano for fast, accurate object detection in real-time applications and resource-constrained environments.
3. Tackle Indian road-specific issues like erratic driving, dense traffic, and diverse objects using a tailored dataset.



Future Scope

1. Ensure high detection accuracy across diverse weather and lighting conditions, including rain, and night-time driving.
2. Build a scalable, efficient system suitable for deployment in autonomous vehicles and other real-time applications.



Dataset Description

1. We used the DATS_2022 dataset, specifically designed for Indian roads, containing:

2. 1,861 training images and 566 validation images

3. 35 distinct classes such as Traffic Signal, Lamp Post, Zebra Crossing, Bike, Car, Rikshaw, Tyre Works, Tree, Tractor, Cattle, Vegetation, Electricity Pole, Building, Board Wall, person, Bus, Bridge, Road Divider, Tempo, Traffic Sign, Board, Flag, Crane, Cycle, Dog, Truck, Overbridge, Manhole, Bus Stop, Barricade, Petrol Pump, Ambulance, Goat, Cart

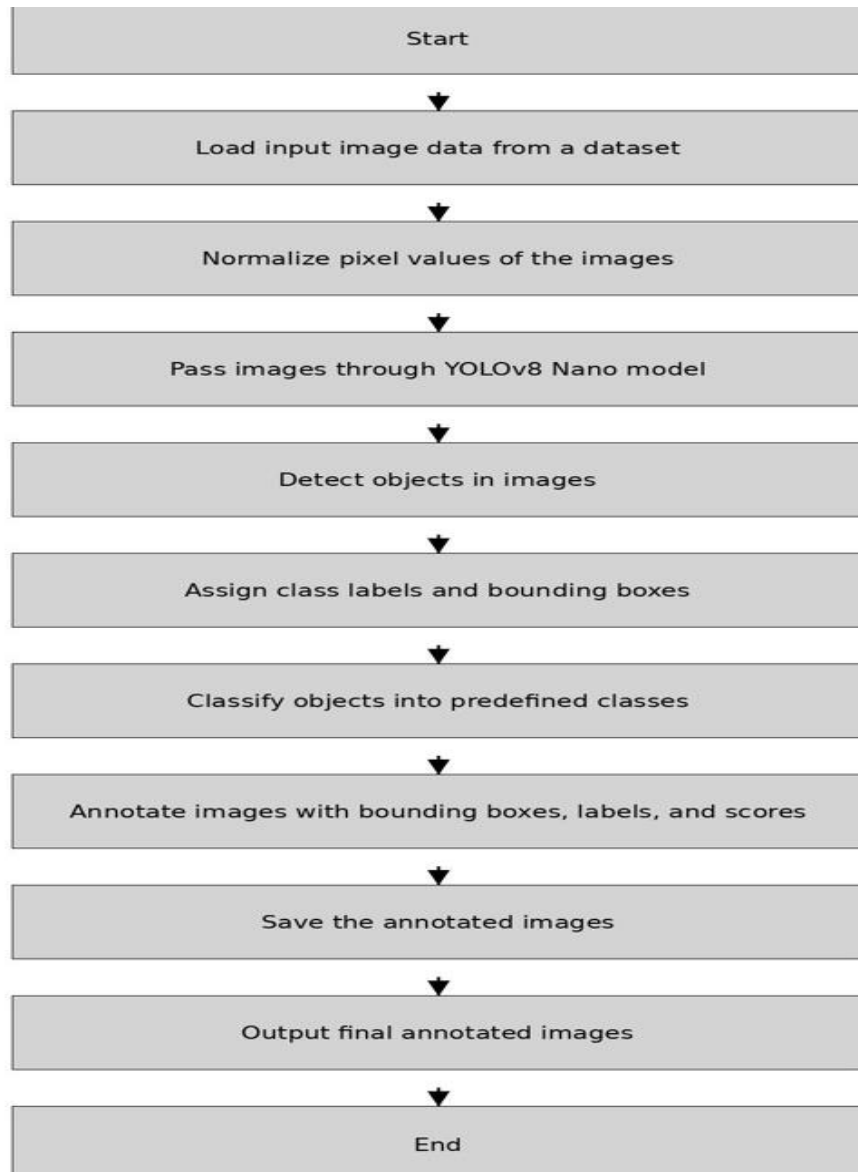
4. The dataset was augmented using techniques such as rotation, scaling, flipping, and brightness adjustments to ensure model robustness across various environmental conditions like rain, fog, and night-time driving.

Dataset-link:

<https://data.mendeley.com/datasets/nfc34n8svj/2>



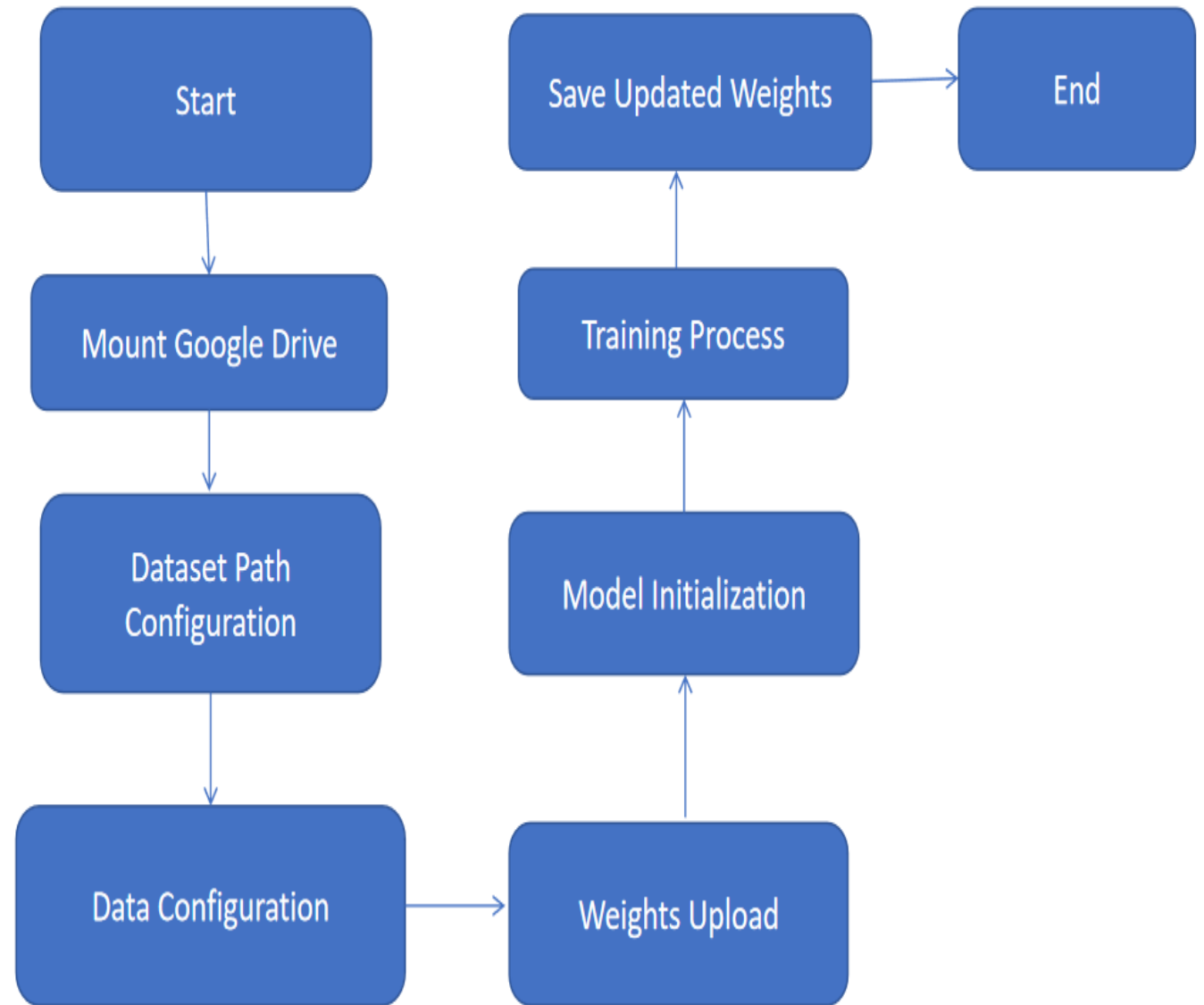
Proposed Methodology



- **Start the Process:** Begin the object detection workflow.
- **Load Image Data:** Load input images from a dataset.
- **Normalize Images:** Scale the pixel values of images for consistency.
- **Pass Through YOLOv8 Nano:** Input the images into the YOLOv8 Nano model for analysis.
- **Detect Objects:** Identify objects present in the images.
- **Assign Labels and Bounding Boxes:** Label the detected objects and define their locations with bounding boxes.
- **Classify Objects:** Categorize the detected objects into predefined classes.
- **Annotate Images:** Add bounding boxes, labels, and confidence scores to the images.
- **Save Annotated Images:** Save the processed images with annotations.
- **Output Final Images:** Provide the final set of annotated images as output.
- **End the Process:** Complete the workflow.

Workflow/ Implement model

Workflow



Results and Discussion

Hyperparameters:

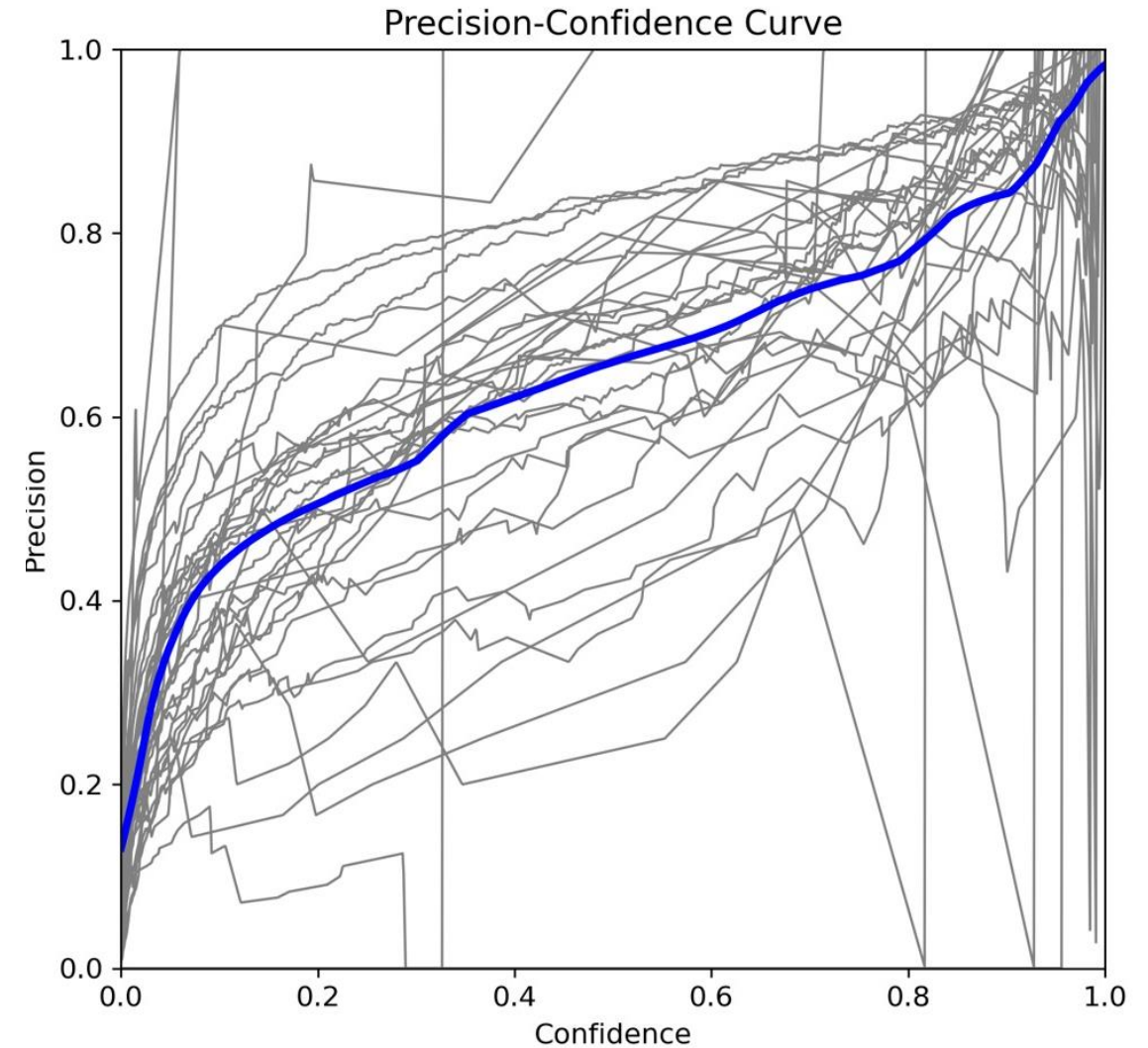
- 1. Learning Rate: Determines the step size for updating model weights during training (e.g., 0.001).
- 2. Batch Size: The number of samples processed before the model updates its weights (e.g., 32 or 64).
- 3. Number of Epochs: The number of times the entire training dataset is passed through the model (e.g., 50).
- 4. Dropout Rate: A fraction of input units to drop during training to prevent overfitting (e.g., 0.3).
- 5. Optimizer: The optimization algorithm used for training (e.g., Adam, SGD, or RMSprop).

Metric	Value (%)
Accuracy	54
Precision	55
Recall (Sensitivity)	46
Specificity	62
F1-score	50
mAP@0.5	54.9

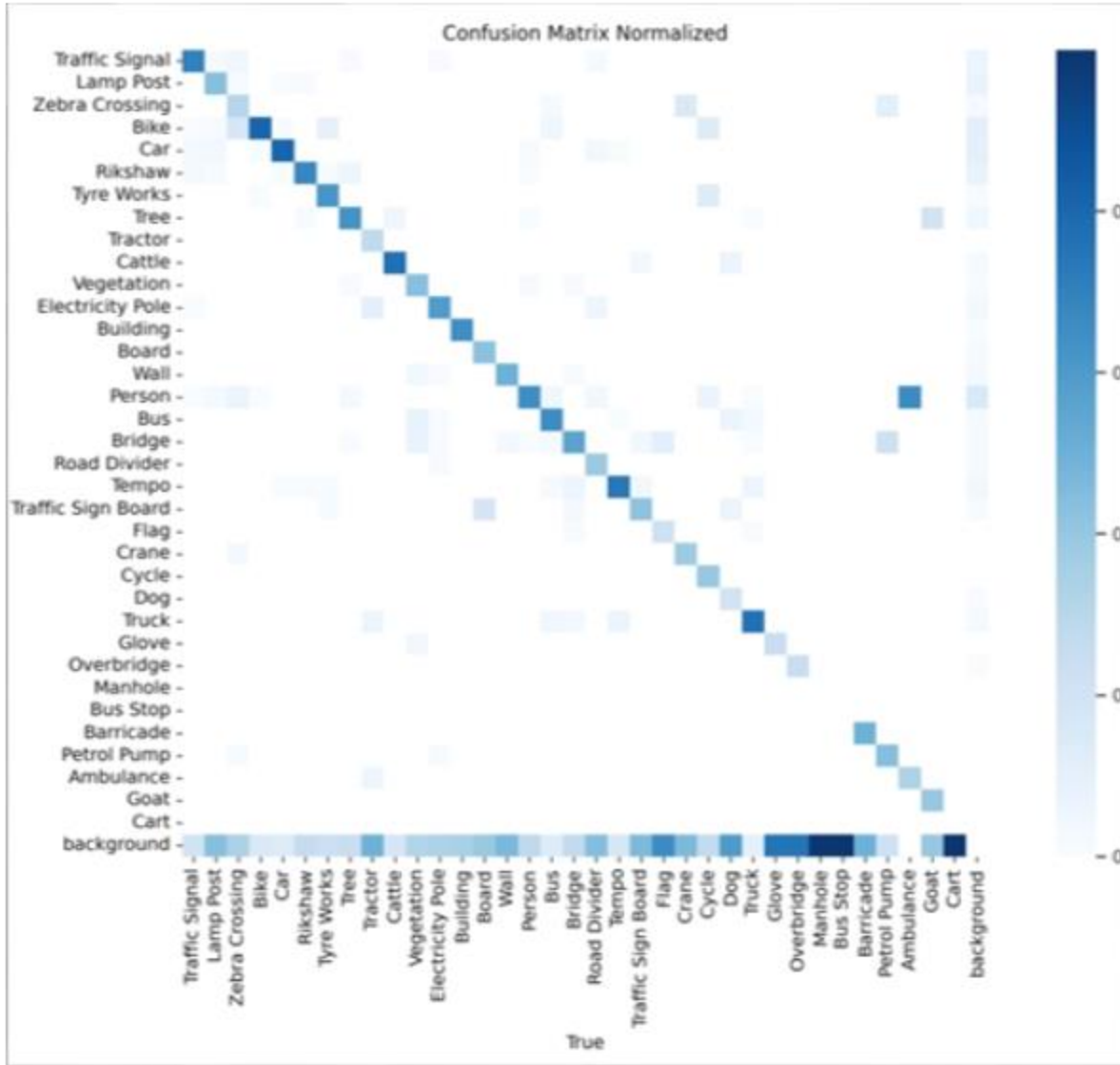
Results and Discussion

Results and Discussion

In the precision-confidence curve reveals an overall precision of 0.98 at a confidence threshold of 1.0, though performance varies across different object classes, highlighting opportunities for further refinement in detecting smaller objects.



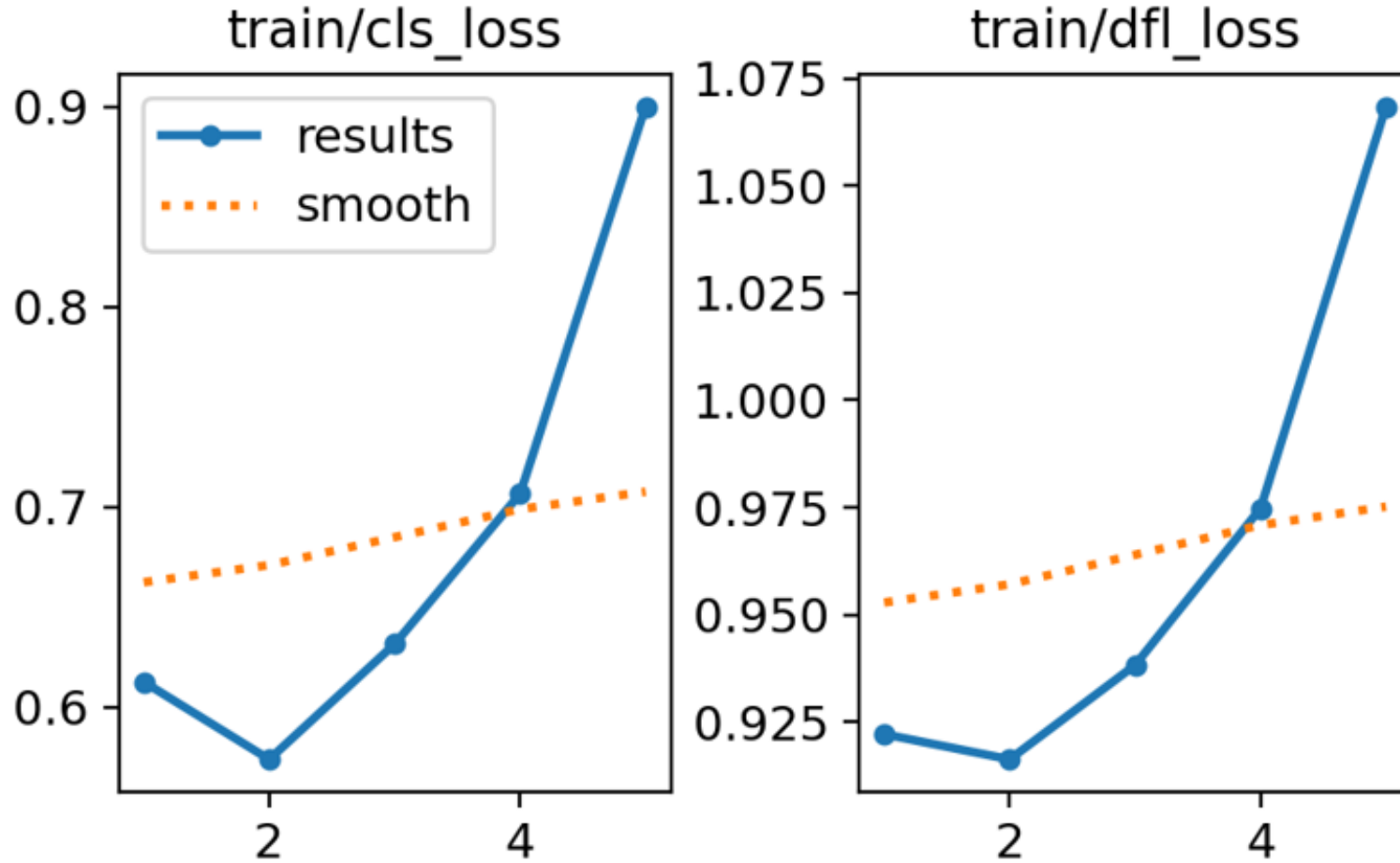
Results and Discussion



- This image represents a **normalized confusion matrix** for a multi-class classification task.
- **Rows:** Predicted classes.
- **Columns:** True classes.
- **Diagonal:** Correct predictions (high values indicate good performance for that class).
- **Off-diagonal values:** Misclassifications (showing which classes were confused with others).
- The color intensity reflects the proportion of predictions, with darker colors indicating higher values.

Results and Discussion

GRAPH PLOT FOR EPOCH NO
VS ERROR COST

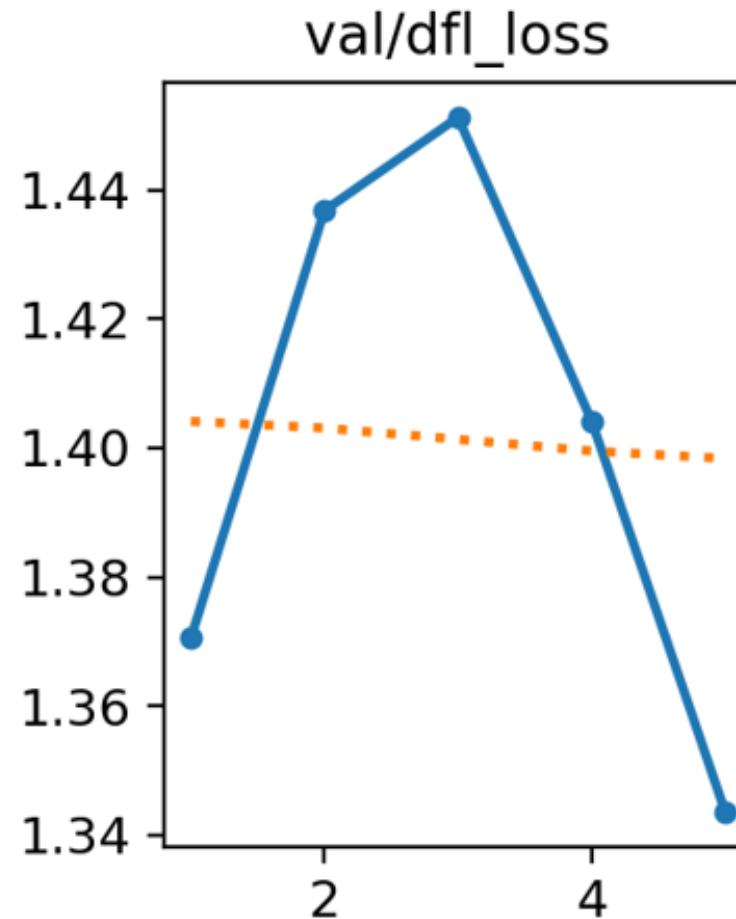
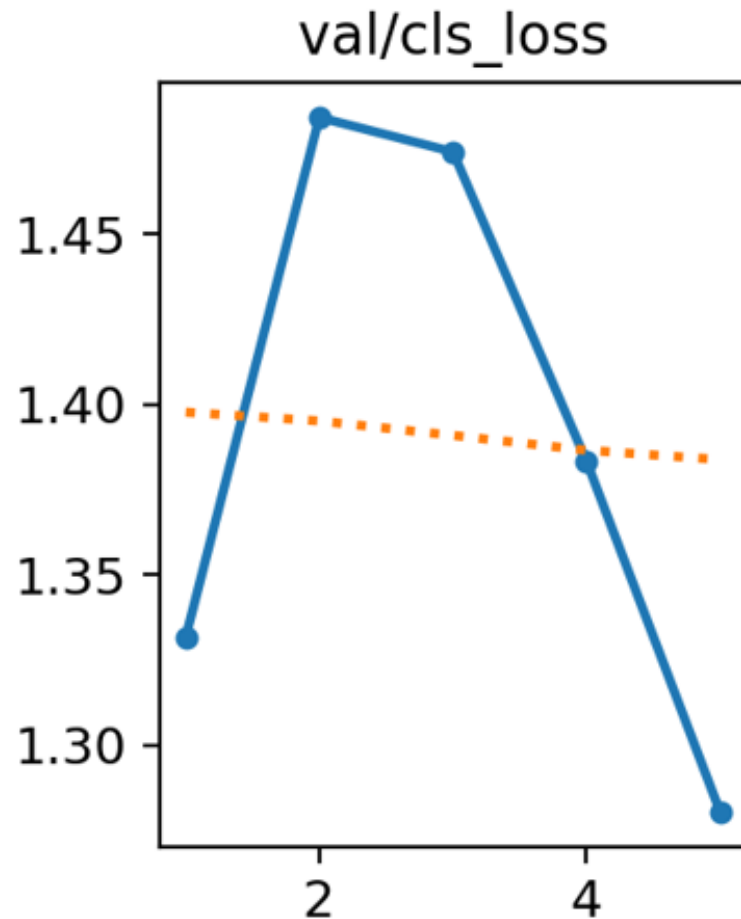


1. Train/cls_loss: The classification loss for the training data initially decreases but then increases sharply, indicating potential overfitting or instability in later epochs.

2. Train/df_l_loss: The distribution-focused loss for training follows a similar pattern, starting low and rising significantly as training progresses.

Results and Discussion

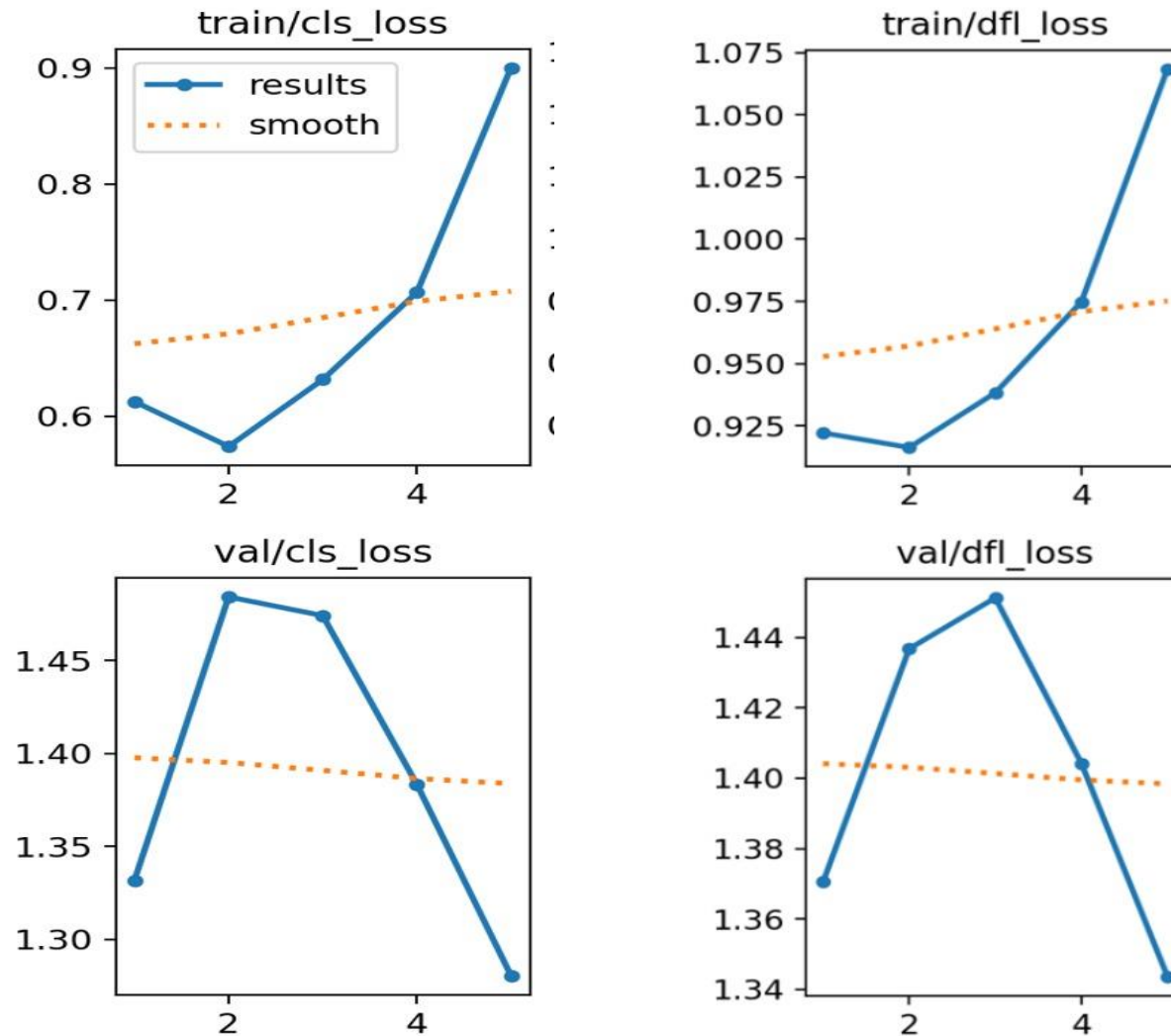
GRAPH PLOT FOR EPOCH NO
VS ERROR COST



1. Val/cls_loss: The classification loss for validation increases initially but then decreases sharply, suggesting the model might be improving its generalization in later epochs.

2. Val/df_l_loss: The distribution-focused loss for validation shows a similar trend, with an initial rise followed by a significant decline, indicating better performance on unseen data in later epochs.

Results and Discussion



GRAPH OF TRAINING ERROR
AND TESTING ERROR

Results and Discussion

Sample input Images:-

DAYLIGHT IMAGE BEFORE DETECTION

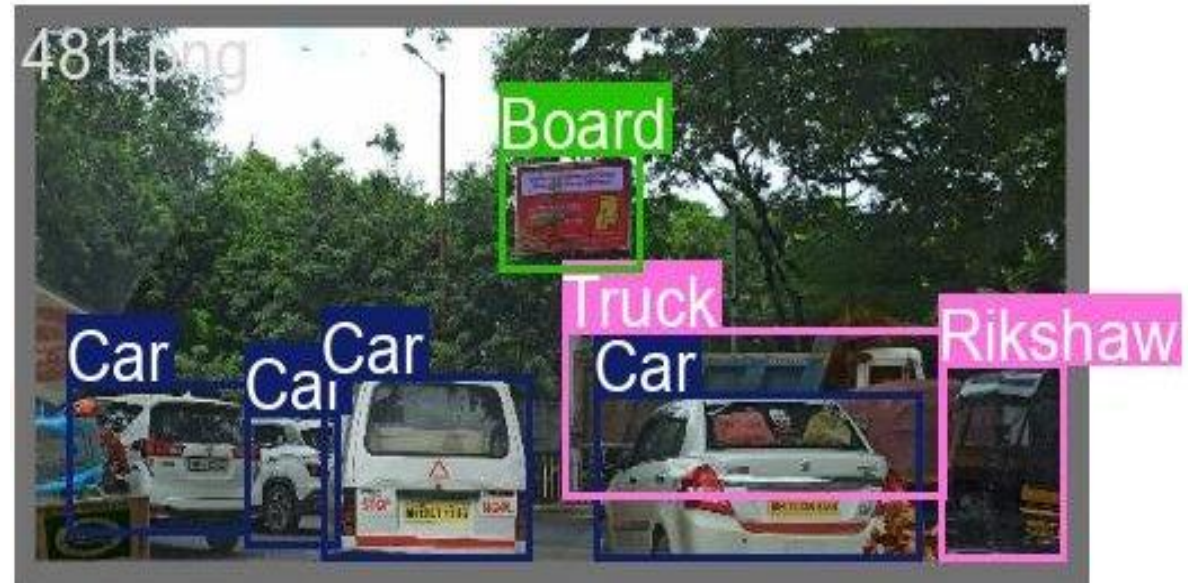


FOGGY IMAGE BEFORE DETECTION



Results and Discussion

DAYLIGHT IMAGE AFTER DETECTION

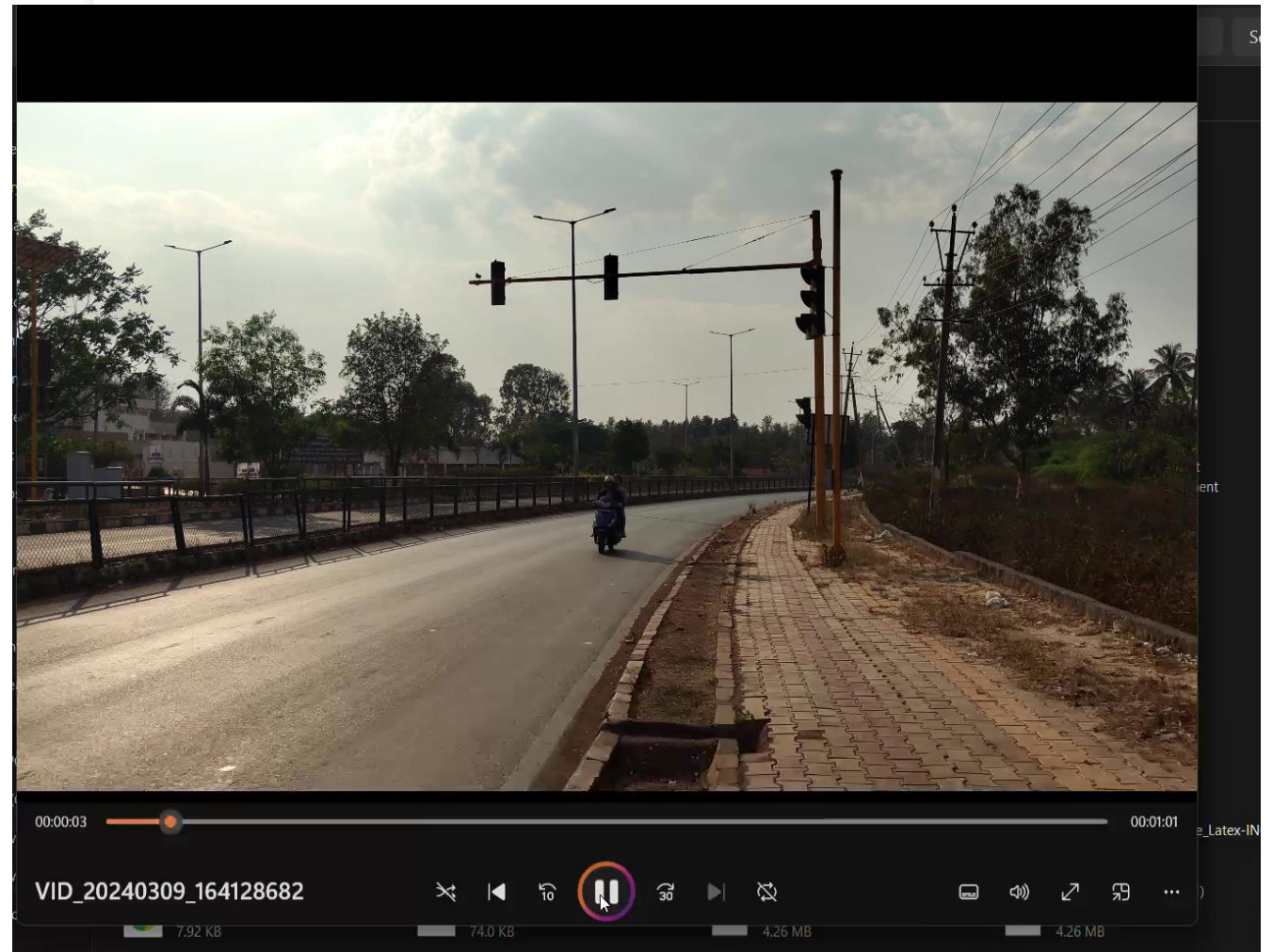


FOGGY IMAGE AFTER DETECTION



Results and Discussion

The video focuses on detecting objects such as vehicles, pedestrians, and road signs and classifying them into categories like cars, buses, bikes, and many more. Using YOLOv8 ensures high accuracy and speed, crucial for real-time analysis. Indian roads present unique challenges, including dense, unstructured traffic, diverse object types, occlusions, and varying lighting or weather conditions. YOLOv8's robustness and efficiency make it ideal for tackling these complexities. The applications include traffic monitoring, road safety analysis, smart transportation systems, and support for autonomous vehicle technologies.



References

- [1] **C. Pimpalgaonkar, S. Shinde, and R. Sharma**, “Lane and Object Detection using YOLO: Indian Roads Scenario,” *Journal of Image Processing and Intelligent Remote Sensing*, vol. 04, no. 04, pp. 23– 29, June-July 2024. DOI: <https://doi.org/10.55529/jipirs.44.23.29>.
- [2] **C. Kamjula , U. Khuteta , and B. K. Tripathy**, “Comparative Analysis of Efficiency of YOLO (v5 and v8) Models to Detect Potholes and Traffic Signs on Indian Roads,” in *6th International Conference on Computational Intelligence in Pattern Recognition (CIPR-2024)*, Tamil Nadu, India, pp. 1–10, March 2024.
- [3] **Y. Fujitake, T. Sugimoto**. "Stochastic Adversarial Video Prediction for Object Detection in Real Time." *Advances in Intelligent Systems and Computing*, Springer, pp. 1041-1052, 2022.
- [4] **Y. Lim, S. S. Tiang, W. H. Lim, C. H. Wong, M. Mokayef, K. S. Chong, and B. Sun**. "Object Detection in Autonomous Vehicles: A Performance Analysis." *Advances in Intelligent Manufacturing and Robotics*, Springer, vol. 845, pp. 277-289, February 2024. DOI: https://doi.org/10.1007/978-981-99-8498-5_21.
- [5] **A. J. Tuaimah, H. B. Taher**. "Detection and Classification of Moving Objects." *Journal of for Pure Science*, University of Thi-Qar , vol. 10, no. 2, pp. 41-51, June 2020. DOI: <http://doi.org/10.32792/utq.jceps.10.02.05>

References

- [6] **A. J. Tuaimah and H. B. Taher**, “Detection and Classification System for Moving Objects in Video Surveillance Using Background Subtraction and K-Nearest Neighbor,” *Journal of Intelligent Systems*, vol. 12, no. 4, pp. 233–242, 2022.
- [7] **Y. Xu, Y. Zhang, L. Zhao, and T. Li**, “Area-Aware Attention Mechanism for Road Abandoned Object Detection Using Video Surveillance,” *IEEE Transactions on Intelligent Transportation Systems*, vol. 23, no. 8, pp. 5432–5443, Aug. 2023.
- [8] **P. Madhan and K. Shanmugapriya**, “Enhanced Moving Object Detection (EMOD) System Using Hyperbolic Shadow Detection and Manta Ray Optimized CNN,” *Springer Lecture Notes in Computer Science*, vol. 1360, pp. 112–126, 2023.
- [9] **E. Abbasi, M. Ebrahimi, and A. Rahmani**, “Hyper Parameterized YOLOv8 for Object and Pedestrian Detection in Foggy Weather Conditions,” *Elsevier Journal of Transportation Safety and Environment*, vol. 17, no. 3, pp. 221–232, 2024.
- [10] **S. Srinath and V. Baydeti**, “Deep Learning-Based Video Classification Using CNN and LSTM for UCF101 Dataset,” in *Proceedings of the ACM Multimedia Conference*, Seoul, South Korea, pp. 89–98, 2023.
- [11] **R. Guo and H. Chen**, “Multi-Object Road Waste Detection and Classification Using Binocular Vision and CNN,” *IEEE Access*, vol. 31, no. 5, pp. 3987–3998, June 2023.

References

- [12] **Adegun, A.A., Fonou-Dombeu, J.V., Viriri, S and Odindi, J.** “Ontology-Based Deep Learning Model for Object Detection and Image Classification in Smart City Concepts,” *Smart Cities*, vol. 7, no. 4, pp. 2182-2207, Aug. 2024. DOI: <https://doi.org/10.3390/smartcities7040086>.
- [13] **K. Ding, Z. Ding, Z. Zhang, M. Yuan, G. Ma, and G. Lv,** "SCD-YOLO: A Novel Object Detection Method for Efficient Road Crack Detection," *Research Article*, June 7, 2024. DOI: <https://doi.org/10.21203/rs.3.rs-4350707/v1>

THANK YOU

